

UNIVERSIDAD PERUANA CAYETANO HEREDIA

FACULTAD DE CIENCIAS E INGENIERÍA

INFORME 2

SISTEMAS OPERATIVOS



Grupo 4

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INFORME 2



ACTIVIDAD

Revisar el material de Comandos Linux de esta sesión respecto a los comandos que tiene que ver con las actividades de los procesos.

Ejemplo:

- Kill
- Killall
- Top
- Htop
- Nice

- **kill:** Finaliza un proceso usando su número identificador, llamado PID en la forma de: **kill (PID)**

```
ubuntu@ubuntu:~/Sistemas-Operativos-$ kill
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sig
spec]
```

```
ubuntu@ubuntu:~$ kill --help
kill: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]
Send a signal to a job.
```

Send the processes identified by PID or JOBSPEC the signal named by SIGSPEC or SIGNUM. If neither SIGSPEC nor SIGNUM is present, then SIGTERM is assumed.

Options:

- s sig SIG is a signal name
- n sig SIG is a signal number
- l list the signal names; if arguments follow '-l' they are assumed to be signal numbers for which names should be listed
- L synonym for -l

Kill is a shell builtin for two reasons: it allows job IDs to be used instead of process IDs, and allows processes to be killed if the limit on processes that you can create is reached.

Exit Status:

Returns success unless an invalid option is given or an error occurs.

- **killall:** Finaliza todos los procesos que tengan un nombre determinado.

```
ubuntu@ubuntu:~$ killall
Usage: killall [OPTION]... [--] NAME...
       killall -l, --list
       killall -v, --version

-e,--exact          require exact match for very long names
-I,--ignore-case    case insensitive process name match
-g,--process-group  kill process group instead of process
-y,--younger-than   kill processes younger than TIME
-o,--older-than     kill processes older than TIME
-i,--interactive    ask for confirmation before killing
-l,--list           list all known signal names
-q,--quiet          don't print complaints
-r,--regex          interpret NAME as an extended regular expression
-s,--signal SIGNAL  send this signal instead of SIGTERM
-u,--user USER      kill only process(es) running as USER
-v,--verbose        report if the signal was successfully sent
-V,--version        display version information
-w,--wait           wait for processes to die
-n,--ns PID         match processes that belong to the same namespaces
                    as PID
-Z,--context REGEXP kill only process(es) having context
                    (must precede other arguments)

ubuntu@ubuntu:~$
```

- **top:** Muestra en tiempo real los procesos activos y el consumo de CPU y memoria.

```
ubuntu@ubuntu:~$ top
top - 19:04:52 up 50 min, 1 user, load average: 0.02, 0.06, 0.07
Tasks: 244 total, 1 running, 243 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.1 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 24444.1 total, 13302.4 free, 1984.6 used, 9901.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 22459.5 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
 3891 ubuntu    20   0 5143288 405228 172092 S  0.7   1.6   1:01.64 gnome-shell
 1642 root       20   0      0      0      0 I  0.3   0.0   0:05.09 kworker/u18:5-events_freezable_pwr_efficient
 7857 root       20   0      0      0      0 I  0.3   0.0   0:00.22 kworker/1:2-events
 7861 ubuntu    20   0 2919804 388760 144920 S  0.3   1.6   0:01.47 ptysix
 7919 ubuntu    20   0 22096  6188  4036 R  0.3   0.0   0:00.02 top
    1 root       20   0  26356 17660 11884 S  0.0   0.1   0:04.62 systemd
    2 root       20   0      0      0      0 S  0.0   0.0   0:00.03 kthreadd
    3 root       20   0      0      0      0 S  0.0   0.0   0:00.00 pool_workqueue_release
    4 root       0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root       0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-sync_wq
    6 root       0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
    7 root       0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-slub_flushwq
    8 root       0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-netns
   10 root       0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/0:0H-kblockd
   13 root       0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
   14 root       20   0      0      0      0 S  0.0   0.0   0:00.12 ksoftirqd/0
   15 root       20   0      0      0      0 I  0.0   0.0   0:01.48 rcu_preempt
   16 root       20   0      0      0      0 S  0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
   17 root       20   0      0      0      0 S  0.0   0.0   0:00.34 rcu_exp_gp_kthread_worker
   18 root       rt    0      0      0      0 S  0.0   0.0   0:00.08 migration/0
   19 root       20   0      0      0      0 S  0.0   0.0   0:00.00 kprobe-optimizer
   20 root      -51   0      0      0      0 S  0.0   0.0   0:00.00 idle_inject/0
   21 root       20   0      0      0      0 S  0.0   0.0   0:00.00 cpuhp/0
   22 root       20   0      0      0      0 S  0.0   0.0   0:00.00 cpuhp/1
   23 root      -51   0      0      0      0 S  0.0   0.0   0:00.00 idle_inject/1
   24 root       rt    0      0      0      0 S  0.0   0.0   0:00.09 migration/1
   25 root       20   0      0      0      0 S  0.0   0.0   0:00.87 ksoftirqd/1
   28 root       20   0      0      0      0 S  0.0   0.0   0:00.00 cpuhp/2
   29 root      -51   0      0      0      0 S  0.0   0.0   0:00.00 idle_inject/2
   30 root       rt    0      0      0      0 S  0.0   0.0   0:00.07 migration/2
   31 root       20   0      0      0      0 S  0.0   0.0   0:00.15 ksoftirqd/2
```

- **htop**: es una alternativa más amigable a top, con una interfaz interactiva y visual

```

0.0%] Tasks: 133, 492 thr, 131 kthr; 3 running
0.7% Load average: 0.01 0.05 0.07
0.0% Uptime: 00:55:36
0.7%
1.99G/23.9G
0K/0K

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
3891 ubuntu 20 0 5074M 456M 199M S 1.3 1.9 0:37.23 /usr/bin/gnome-shell --mode=ubuntu
3981 ubuntu 20 0 5078M 456M 0 S 0.7 1.9 0:08.33 /usr/bin/gnome-shell --mode=ubuntu
7861 ubuntu 20 0 2859M 388M 145M S 0.7 1.6 0:02.21 /usr/bin/ptxix --gaplication-service
7886 ubuntu 20 0 2859M 388M 0 S 0.7 1.6 0:00.65 /usr/bin/ptxix --gaplication-service
7887 ubuntu 20 0 2859M 388M 0 S 0.7 1.6 0:00.70 /usr/bin/ptxix --gaplication-service
8636 ubuntu 20 0 20728 5644 3796 R 0.7 0.0 0:00.11 htop
1 root 20 0 26356 17660 11884 S 0.0 0.1 0:04.65 /sbin/init autoinstall --- splash noprompt noshell automatic-ubiquity
967 root 19 -1 50972 20000 18168 S 0.0 0.1 0:00.86 /usr/lib/systemd/systemd-journald
1021 systemd-oo 20 0 16808 7728 6664 S 0.0 0.0 0:00.31 /usr/lib/systemd/systemd-oomd
1025 root 20 0 41560 15144 9156 S 0.0 0.1 0:00.55 /usr/lib/systemd/systemd-udev
1027 systemd-re 20 0 22672 15132 12384 S 0.0 0.1 0:00.24 /usr/lib/systemd/systemd-resolved
2002 avahi 20 0 6744 4712 4324 S 0.0 0.0 0:00.19 avahi-daemon: running [ubuntu.local]
2003 root 20 0 2888 2032 1916 S 0.0 0.0 0:00.08 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
2004 messagebus 20 0 11752 8440 4676 S 0.0 0.0 0:01.62 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation
2008 root 20 0 58112 30112 14836 S 0.0 0.1 0:01.19 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
2009 polkitd 20 0 302M 11288 8068 S 0.0 0.0 0:00.44 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
2012 root 20 0 2050M 46988 29188 S 0.0 0.2 0:01.78 /usr/lib/snapd/snapd
2014 root 20 0 312M 8184 7484 S 0.0 0.0 0:00.25 /usr/libexec/accounts-daemon
2015 root 20 0 18344 3228 2992 S 0.0 0.0 0:00.04 /usr/sbin/cron -F -P
2018 root 20 0 311M 7060 7080 S 0.0 0.0 0:00.07 /usr/libexec/switcheroo-control
2022 root 20 0 18604 9732 8300 S 0.0 0.0 0:00.44 /usr/lib/systemd/systemd-logind
2026 root 20 0 165M 7172 5724 S 0.0 0.0 0:00.02 /usr/sbin/zed -F
2030 avahi 20 0 6560 1700 1396 S 0.0 0.0 0:00.06 avahi-daemon: chroot helper
2039 root 20 0 165M 7172 0 S 0.0 0.0 0:00.00 /usr/sbin/zed -F
2040 root 20 0 165M 7172 0 S 0.0 0.0 0:00.00 /usr/sbin/zed -F
2059 root 20 0 311M 7860 0 S 0.0 0.0 0:00.00 /usr/libexec/switcheroo-control
2060 root 20 0 311M 7860 0 S 0.0 0.0 0:00.00 /usr/libexec/switcheroo-control
2066 root 20 0 312M 8184 0 S 0.0 0.0 0:00.00 /usr/libexec/accounts-daemon
1 Help F2 Setup F3 Search F4 Filter F5 Free F6 SortBy F7 Nice F8 Nice F9 Kill F10 Quit

```

- **nice**: permite iniciar un programa con una determinada prioridad de CPU en la forma de: **nice -n (valor) (programa)**

```

vboxuser@UbuntuNya:~$ nice -n 10 firefox
update.go:193: cannot change mount namespace according to change mount (/var/lib/snapd/hostfs/usr/local/share/doc /usr/local/share/doc none bind,ro
hare/doc" because it would affect the host in "/var/lib/snapd"
update.go:193: cannot change mount namespace according to change mount (/var/lib/snapd/hostfs/usr/share/gimp/2.0/help /usr/share/gimp/2.0/help
/share/gimp/2.0/help" because it would affect the host in "/var/lib/snapd"
update.go:193: cannot change mount namespace according to change mount (/var/lib/snapd/hostfs/usr/share/gtk-doc /usr/share/gtk-doc none bind,ro
oc" because it would affect the host in "/var/lib/snapd"
update.go:193: cannot change mount namespace according to change mount (/var/lib/snapd/hostfs/usr/share/libreoffice/help /usr/share/libreoffice
fs/usr/share/libreoffice/help" because it would affect the host in "/var/lib/snapd"
update.go:193: cannot change mount namespace according to change mount (/var/lib/snapd/hostfs/usr/share/sphinx_rtd_theme /usr/share/sphinx_rtd_
fs/usr/share/sphinx_rtd_theme" because it would affect the host in "/var/lib/snapd"
update.go:193: cannot change mount namespace according to change mount (/var/lib/snapd/hostfs/usr/share/xubuntu-docs /usr/share/xubuntu-docs no
hare/xubuntu-docs" because it would affect the host in "/var/lib/snapd"
libpbackend-1.0.so: cannot open shared object file: No such file or directory
Failed to load module: /home/vboxuser/snap/firefox/common/.cache/gio-modules/libgiolibproxy.so
Gtk-Message: 19:18:27.533: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
nice
[Parent 6828, Main Thread] WARNING: Couldn't map window 0x7d3cd9581900 as subsurface because its parent is not mapped.: 'glib warning', file /b
p:201
(firefox_firefox:6828): Gdk-WARNING **: 19:19:17.707: Couldn't map window 0x7d3cd9581900 as subsurface because its parent is not mapped.
vboxuser@UbuntuNya:~$ nice
0
vboxuser@UbuntuNya:~$

```

```

ubuntu@ubuntu:~/Sistemas-Operativos$ Nice
Command 'Nice' not found, did you mean:
  command 'nice' from deb coreutils-from-gnu (0.0.0~ubuntu25)
  command 'nice' from deb coreutils-from-uutils (0.0.0~ubuntu25)
  command 'nice' from deb coreutils-from-toybox (0.0.0~ubuntu25)
  command 'rice' from deb golang-rice (1.0.2-1build2)
Try: sudo apt install <deb name>

```